

Types of Teams in Cybersecurity

Cybersecurity involves multiple specialized teams and functions. Each contributes a different perspective to protecting systems, applications, users, and data. Team names and responsibilities can vary between organizations; this color-based model is useful for understanding the major security perspectives.

1. Overview of Cybersecurity Teams

Team	Primary Focus	Main Goal
Red Team	Offensive security	Identify weaknesses through authorized attack simulation.
Blue Team	Defensive security	Detect, prevent, and respond to threats.
Purple Team	Red + Blue collaboration	Improve security through shared learning and feedback.
Yellow Team	Development / secure building	Build and maintain secure applications and systems.
White Team	Governance / coordination	Plan, coordinate, and oversee security activities.
Green Team	Security awareness	Educate users and reduce human-related security risks.

2. Red Team — Offensive Security

The Red Team simulates attacker behavior in an authorized environment to identify weaknesses and evaluate security controls.

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Red Team

Role: Offensive security / ethical hacking

Goal: Identify weaknesses and test the effectiveness of security defenses.

Activity	Example
Penetration Testing	Testing authorized systems for exploitable weaknesses.
Phishing Simulation	Sending controlled simulated phishing messages to evaluate awareness.
Password Testing	Performing authorized password-strength assessments using dictionary or brute-force techniques.
Vulnerability Scanning	Identifying known weaknesses such as outdated software.
Social Engineering	Conducting authorized simulations to evaluate responses to deceptive requests.

3. Blue Team — Defensive Security

The Blue Team focuses on protecting systems, detecting suspicious activity, investigating events, and responding to incidents.

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Blue Team

Role: Defensive security / security operations

Goal: Defend systems, detect threats, and respond to security incidents.

Activity	Example
Network Monitoring	Monitoring traffic for unusual activity or repeated suspicious login attempts.
Incident Response	Investigating and containing an infected system after a malware incident.
Log Analysis	Reviewing logs for patterns such as repeated failed authentication attempts.
Firewall & Security Configuration	Configuring controls to restrict unauthorized traffic or access.
Threat Detection	Using IDS/IPS and other tools to detect potentially malicious activity.

4. Purple Team — Collaboration

The Purple Team represents collaboration between offensive and defensive security functions. In practice, purple teaming is often a collaborative approach rather than a separate permanent team.

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Purple Team

Role: Red + Blue team collaboration

Goal: Improve security through collaboration, feedback, and continuous learning.

Activity	Example
Attack Simulation Review	Reviewing how an authorized attack was performed and how effectively it was detected and handled.
Security Improvement	Using attack findings to improve detection rules, monitoring, and controls.
Joint Exercises	Conducting coordinated attack-and-defense exercises.

5. Yellow Team — Development & Secure Building

The Yellow Team focuses on building applications and systems with security in mind. In this learning model, developers work with security functions to reduce weaknesses during design and development.

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Yellow Team

Role: Developers / builders

Goal: Build applications and systems with security incorporated into development.

Activity	Example
Secure Coding	Writing code that validates input and helps prevent vulnerabilities such as SQL injection.
Fixing Vulnerabilities	Patching or correcting a weakness identified during security testing.
Application Development	Designing secure application features such as robust authentication.

6. White Team — Governance & Management

The White Team provides governance, coordination, and oversight. In security exercises, it may define scope, rules of engagement, objectives, and coordination requirements.

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White Team

Role: Management / governance

Goal: Plan, coordinate, govern, and oversee security activities.

Activity	Example
Policy Creation	Defining security policies and requirements.
Risk Management	Evaluating cybersecurity risks and determining appropriate treatment.
Compliance	Helping the organization meet applicable laws, regulations, standards, and internal requirements.

7. Green Team — Security Awareness

The Green Team focuses on security awareness and education. Its activities help users recognize threats and follow secure practices.

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Green Team

Role: Security awareness / training

Goal: Educate users and strengthen the human side of cybersecurity.

Activity	Example
Security Training	Teaching employees how to recognize phishing and follow secure practices.
Awareness Campaigns	Using communications and educational material to promote security habits.
Phishing Drills	Running controlled phishing simulations to evaluate and improve awareness.

8. How the Teams Work Together

Different teams provide different perspectives, and collaboration helps organizations continuously improve security.

Team	Primary Contribution	Common Collaboration
Red	Attack simulation and security findings	Blue, Purple, White
Blue	Detection, monitoring, response, and defense	Red, Purple, Yellow
Purple	Feedback between attack and defense	Red and Blue
Yellow	Secure development and remediation	Blue, Red, White
White	Governance, scope, policies, and risk oversight	All security functions
Green	Training and security awareness	Employees and security teams

Key Takeaway

Red tests from an attacker's perspective. Blue protects and responds. Purple connects offensive findings with defensive improvements. Yellow builds secure systems. White provides governance and coordination. Green improves security awareness.

Important Note

The color-based team model is a learning framework. Organizations may combine these functions, use different names, or assign responsibilities differently.

Conclusion

Understanding these teams helps explain how offensive testing, defensive operations, secure development, governance, and security awareness fit together within a broader cybersecurity

program.